Determination of bisphenol a and bisphenol B residues in canned peeled tomatoes by reversed-phase liquid chromatography.

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Bisphenol A (BPA) and bisphenol B (BPB) concentrations were determined in peeled canned tomatoes of different brands bought in Italian supermarkets. Tomato samples analyzed were packaged in cans coated with either epoxyphenolic lacquer or low BADGE enamel. A solid phase extraction (SPE) was performed on C-18 Strata E cartridge followed by a step on Florisil cartridge. Detection and quantitation were performed by a reversed phase high-performance liquid chromatography (RP-HPLC) method with both UV and fluorescence detection (FD). On the total of 42 tested tomato samples, BPA was detected in 22 samples (52.4%), while BPB was detected in 9 samples (21.4%). BPA and BPB were simultaneously present in 8 of the analyzed samples. The levels of BPA found in this study are much lower than the European Union migration limits of 3 mg/kg food and reasonably unable to produce a daily intake exceeding the limit of 0.05 mg/kg body weight, established by European Food Safety Authority.